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$$J_{ij} = \frac{\partial \tilde{O}_i}{\partial \theta_j}. \quad (3)$$

Higher $\sigma_{\min}(J)$ and lower $\kappa(J) = \sigma_{\max}/\sigma_{\min}$ are favorable. At $k = 0$, $f(r)$ is Schwarzschild for every w_q , so the w_q Jacobian column vanishes exactly. Nonzero rank away from that boundary does not by itself imply practical recoverability at finite precision. The rank-aware historical benchmark that motivated the mature physical calculation is retained in Appendix A and Fig. 13.

3 Physical Geodesic Shooting and Numerical Validation

The emitter follows the equatorial timelike Hamiltonian

$$H = \frac{1}{2}[-E^2/f + f p_r^2 + L^2/r^2] = -\frac{1}{2}. \quad (4)$$

Its physical coordinate azimuth begins at apocentre: $\phi = \pi$, $r = r_a$, $p_r = t = \tau = 0$. Coordinate ϕ is not treated as a radial anomaly; turning points are detected from $p_r = 0$ and the radial azimuthal period includes apsidal advance.

Photons are propagated in Cartesian spatial coordinates with covariant momentum $\mathbf{p}$ and $p_t = -1$,

$$H_\gamma = \frac{1}{2}[-1/f + \mathbf{p}^2 + (f-1)(\mathbf{n} \cdot \mathbf{p})^2] = 0. \quad (5)$$

Two launch angles are solved so the direct branch reaches the observer plane. The implementation reports failed phases rather than interpolating them or substituting synthetic geodesic proxies. Photon momenta are projected onto the static observer tetrad before sky angles are formed. Redshift uses $1+z = \omega_{\text{emit}}/\omega_{\text{obs}}$, and direct arrival time is compared with $\int (1+z) d\tau_{\text{emit}}$ after removing one common additive constant.

In the Schwarzschild benchmark, all 161 phases succeed for both tested nominal w_q values, which agree because $k = 0$. The independent radial benchmark validates the turning-point period and apsidal advance. The physical 121-point grid also preserves the Hamiltonian constraints and exact rank-loss boundary. Figure 1 shows the arrival-time consistency check.

Figure 2 makes the direct-branch geometry explicit using actual validated emitter states, launch angles, photon integrations, and observer geometry. It is a controlled benchmark visualization, not a radiative-transfer or detector simulation.

4 Physical Observable Complementarity

A large response can reveal that the spacetime changed without revealing which parameter caused the change.

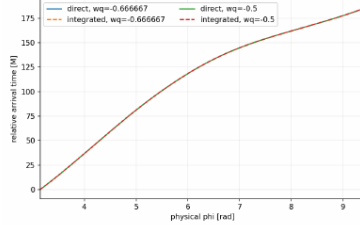


Figure 1: Schwarzschild shooting validation. Physical phase begins at $\phi = \pi$ at apocentre. Direct arrival time and the integrated-redshift curve share only the first-phase additive constant.

Timing and redshift responses are often large but nearly collinear in parameter space. Photon geometry rotates the sensitivity direction and therefore contributes information that ringdown lacks. Figure 3 first shows how the same three validated spacetimes produce distinct phase-resolved response patterns.

The standardized local comparison is shown in Fig. 4. The full observer tracks nearly overlap, as expected for these small deformations. Figure 5 therefore retains the physical track as the primary object and shows unnormalized, pointwise matched-phase residuals relative to Schwarzschild. These small resolved changes supply a direction that is difficult to see on the full track.

Across the accepted physical grid, ringdown plus photon geometry improves the median $\sigma_{\min}$ by 4.5429 $\times$ and the median condition number by 2.2819 $\times$ relative to ringdown alone. The medians for the combined set are $\sigma_{\min} = 1.2326$ and $\kappa(J) = 4.0731$. Ringdown plus all shooting has $\sigma_{\min} = 1.5312$ and $\kappa(J) = 4.4974$: adding more responsive summaries can raise $\sigma_{\min}$ without producing the best conditioning. Figures 6 and 7 show the pointwise maps and their grid-wide summaries, respectively.

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Exact $k = 0$ rank loss is structural, while small nonzero k produces practical ill-conditioning at finite precision. The historical proxy experiment suggested that an additional observable family might improve the weakest parameter direction; this motivated replacing those proxies with validated physical shooting observables.

The third panel is therefore historical motivation rather than evidence from the mature physical solver. The physical results below use timelike-emitter and three-dimensional null-geodesic integrations without proxy substitution.

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Table 1: Primary physical-complementarity summaries. The reported combined-set gains are the archived grid-wide improvement statistics relative to ringdown, not ratios formed from separately reported medians.

Features	med. $\sigma_{\min}$	med. $\kappa(J)$	gain
Ringdown	0.2782	6.6112	reference
Rdown + photon geom.	1.2326	4.0731	4.54 $\times$ / 2.28 $\times$
Rdown + all shooting	1.5312	4.4974	—

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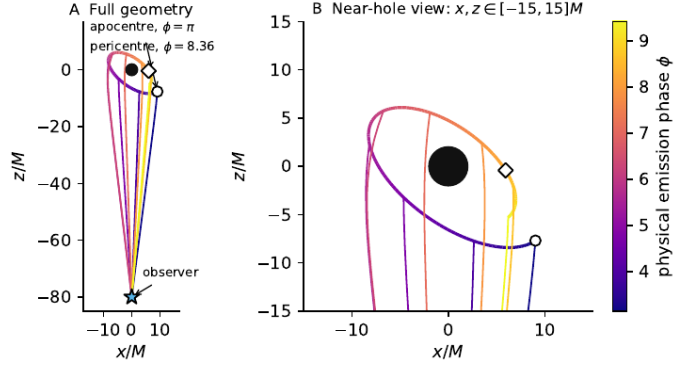


Figure 2: Representative direct-branch null geodesics for $M = 1$, $r_p = 8M$, $r_a = 12M$, $k = 10^{-3}$, $w_q = -0.5$, and observer $(0, 0, -80M)$. Physical phase runs from π at apocentre to 3π ; the diamond marks the numerically detected pericentre and the dark disk is a horizon-scale marker, not a shadow. At each phase, two launch angles are solved until the ray reaches the observer. The right panel shows the stated near-hole coordinate bounds on their actual scale; no trajectory is geometrically magnified. This is a controlled dimensionless benchmark.

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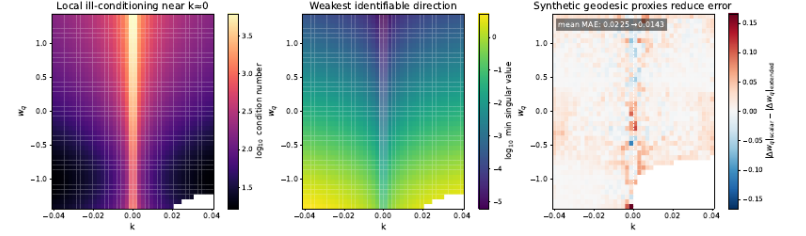


Figure 1: Historical analytic identifiability lens motivating the physical-shooting study. Left: local Jacobian condition number, which grows near $k = 0$. Center: minimum singular value, showing the weakest identifiable direction. Right: change in identifiable- w_q error after adding the historical synthetic geodesic proxies; the mean MAE changes from 0.0225 to 0.0143. White regions are invalid or missing source-grid points and are not interpolated. The first two panels diagnose inverse-map geometry; the third records the original proxy experiment and must not be interpreted as physical ray tracing or geodesic shooting. Proxy augmentation does not resolve the exact $k = 0$ rank loss.

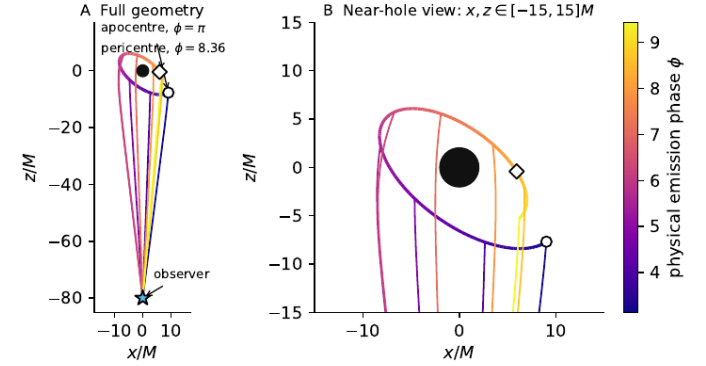


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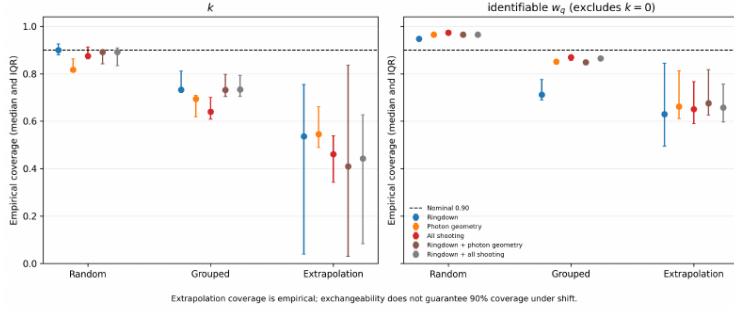


Figure 10: Empirical 90% split-conformal coverage by target and protocol. Ordinary w_q coverage excludes $k = 0$; extrapolation coverage is not guaranteed under distribution shift.

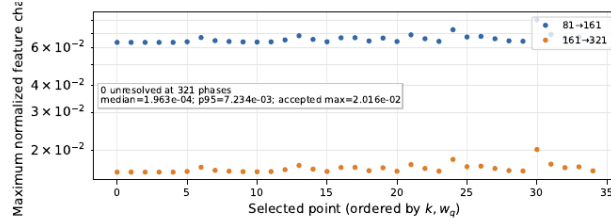


Figure 11: Targeted pointwise maximum normalized feature changes from 81 to 161 and 161 to 321 phases. The 321-phase audit resolves the forward-feature criterion but does not by itself certify inverse estimators.

tary observables, partition by physical systems, test directional shift, and perturb frozen estimators at independently measured numerical scales.

9 Limitations

The ringdown model is leading-eikonal rather than a dedicated finite- ℓ perturbative spectrum. The spacetime is static. Shooting includes only the direct photon branch, a point emitter and point observer, and no radiative transfer, detector response, secondary images, strain likelihood, calibration error, or model discrepancy. The physical grid is finite. Historical proxy results remain synthetic. Although the targeted audit resolves higher-harmonic forward differences on its selected set, it is not a uniform 321-phase grid and tree tail sensitivity persists. MLP extrapolation is unstable. The verdict is therefore conditional, and no result is an observational constraint.

10 Conclusion

Physical complementarity is supported: ringdown constrains k , while photon geometry supplies an independent direction needed for much of the w_q recovery. Random interpolation overstates recoverability relative to grouped and directional validation. Finally, numerical and distribution-shift audits are necessary before confident inverse claims: the forward features can converge while fitted estimators remain fragile. These conclusions motivate robust inverse methods and broader physical modeling before observational use.

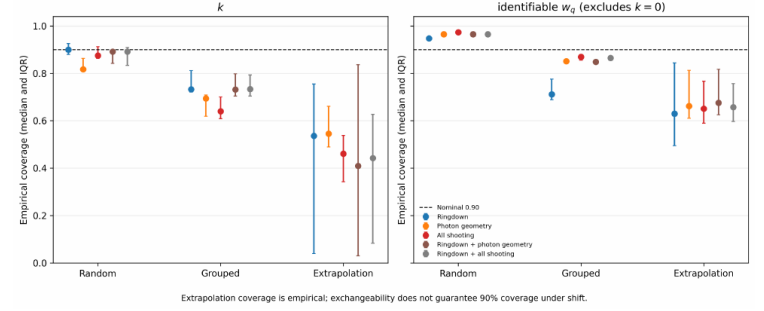


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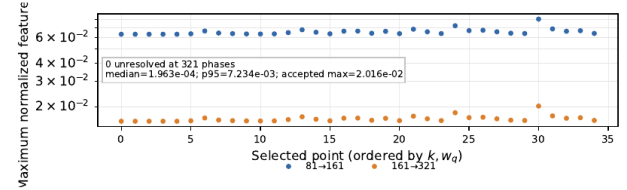


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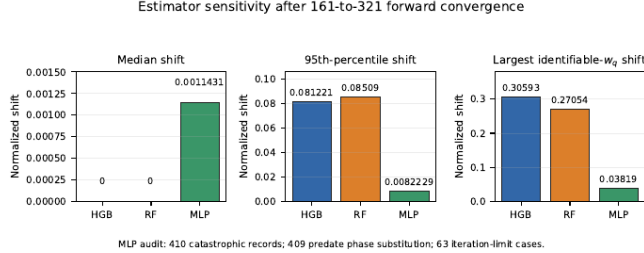


Figure 12: Frozen-estimator sensitivity after forward convergence. Bars show the median, 95th-percentile, and largest identifiable- w_q normalized prediction shift for HGB, RF, and MLP. The annotation records the distinct MLP failure audit: 410 catastrophic directional records, 409 already catastrophic at 161 phases, and 63 iteration-limit cases. No predictions are clipped.

A Historical Synthetic and Proxy Benchmark

The original dense benchmark used leading-eikonal curves, training-fold PCA, and synthetic geodesic proxies. Waveform-only RF achieved $R^2(k) = 0.8960$ and $R^2(w_q) = 0.9587$; HGB with proxy augmentation achieved 0.9793 and 0.9891. The rank-aware classifier reached macro F1 0.9928 with a 57.6% weak-identifiability fraction. PCA compressed correlated information; it did not create a new physical response direction. These historical results motivated the physical shooting study and are not substituted for it.

B Full Split Definitions

Random splits partition parameter points before model-level realization. Grouped physical interpolation holds out blocks under both registered 3×3 layouts. Directional extrapolation separately holds high k , low k , high w_q , and low w_q sides. Calibration points are disjoint from training and test points, and the archived assignments are reused exactly.

C Additional Model-Level Results

HGB and RF reproduce the central grouped photon-geometry improvement for identifiable w_q . The MLP is retained as a stress test rather than omitted: its catastrophic errors are un-clipped and included in aggregate summaries.

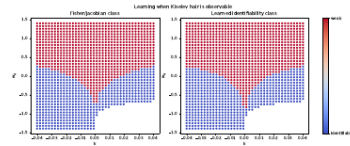


Figure 13: Historical dense synthetic rank-aware identifiability boundary. Exact $k = 0$ rank loss is structural, whereas the finite-width weak-identifiability region depends on precision. This historical benchmark motivated, but does not replace, the validated physical shooting analysis.

D Four Directional Extrapolation Tests

Directions are stored independently in the prediction tables before any median summary. High-to-low and low-to-high tests differ in training support and cannot be interpreted as interchangeable folds.

E Conformal Calibration and Rejection

Split-conformal residual quantiles are fitted on calibration subsets. The exchangeability guarantee applies to matching distributions; shifted coverage is reported empirically. Rejection curves rank uncertainty but do not repair distribution shift.

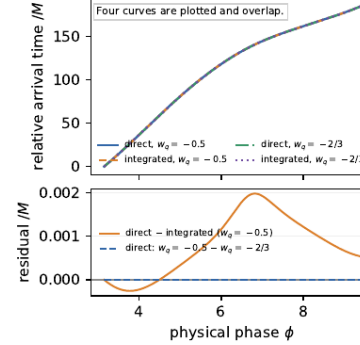


Figure 14: Schwarzschild arrival-time validation. Four arrival-time curves are plotted. The two nominal w_q cases coincide because the $k = 0$ geometry is Schwarzschild, while the direct and integrated-redshift constructions agree within numerical quadrature error. The lower panel shows direct minus integrated arrival time for $w_q = -0.5$ and the direct-arrival difference between $w_q = -0.5$ and $w_q = -2/3$ on their actual numerical scales.

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F Noise and Learning Curves

Grouped one-percent feature noise yields ringdown-plus-photon-geometry NMAE 0.0699/0.0853 for k/w_q . Learning curves are still improving at 78 training points, so data limitation remains part of the interpretation.

G Numerical-Validation Thresholds

The 35-point 321-phase run takes 5318.2 s. Maximum hit, timelike-constraint, null-constraint, and impact-parameter drift errors are respectively 7.0913×10^{-8} , 3.1303×10^{-13} , 7.6214×10^{-12} , and 1.7932×10^{-11} . The Schwarzschild arrival-time comparison in Fig. 14 separates the visually overlapping curves from their actual-scale residuals.

Matched Schwarzschild and Kiselev direct rays are compared on their actual coordinate scale in Fig. 15.

Table 2: Targeted 321-phase frozen-estimator robustness.

Model	median	95th percentile	max. w_q
HGB	0	0.08122	0.30593
RF	0	0.08509	0.27053
MLP	0.001143	0.008223	0.03819

H Uniform 161-Phase Convergence Audit

All 121 points complete in 4473.2 s with zero failed points. The audit preserves the exact $k = 0$ boundary and uses no mixed-resolution row, interpolation, or proxy substitution.

I MLP Extrapolation-Failure Audit

The machine-readable table contains standardized input magnitude, hidden activation magnitude, iteration status, overflow flag, training distance, and resolution response for every catastrophic record. It distinguishes optimization failure from extrapolation instability without clipping.

Exact frozen-estimator shift summaries used in Fig. 12 are retained in Table 2 for numerical reference.